

UJJALDEEP SINGH

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EDUCATION

Indian Institute of Information Technology, Una
B.Tech CSE (Cyber Security); GPA: 8.39 | Branch Silver Medalist

July 2024 – June 2028
Himachal Pradesh, India

Gurukul The School
Class XII; 97.40%

April 2024
Ghaziabad, U.P.

EXPERIENCE

**Department for Promotion of Industry and Internal Trade,
Ministry of Commerce and Industry, Government of India**
Software Development Intern

June 2025 – July 2025
New Delhi, India

- Engineered Veridian, a Flask-based password manager supporting encrypted storage of 1k+ credentials.
- Architected authentication workflows with database integration, improving request handling efficiency by 35%.
- Devised AES-based encryption for local storage, ensuring secure protection of all sensitive user data at rest.
- Automated deployment pipelines using Batch and PowerShell scripts, reducing setup time by 60%.
- Optimized frontend interactions, reducing user errors by 25% and improving overall usability.

PROJECTS

NuanceNode – Structured Claim Analysis System | GitHub | Live Demo
Python, FastAPI, Supabase, React.js, REST APIs, Ollama, Redis

March 2026 – Present

- Architected a multi-stage claim analysis pipeline (facts, sources, logic, belief, verdict), enabling structured reasoning across 5 stages.
- Built scalable REST APIs handling 1k+ requests/day with sub-300ms latency for non-AI endpoints.
- Integrated local LLM inference via Ollama, orchestrating 5+ prompt stages, improving response consistency by 30% and reducing token usage by 25%.
- Engineered data layer using SQLAlchemy with MySQL/Postgres, supporting 10k+ records with <100ms query time.
- Enhanced UX with staged loaders, reducing perceived latency by 35% and improving session completion rate by 20%.

NullPass – Passwordless Auth System | GitHub | Live Demo
Python, Django, Cryptography, REST APIs, PostgreSQL

February 2026

- Implemented ECDSA-based challenge-response authentication, eliminating passwords/OTPs and reducing attack surface by 70%.
- Developed secure Django APIs for key registration and signature verification, handling 500+ requests/day with <200ms latency.
- Established device-bound cryptographic identity, reducing credential theft and replay attack risk by 60%.
- Built full-stack authentication flow (React + Django) with client-side key generation, ensuring zero private key exposure.
- Designed zero shared-secret architecture, improving resilience against database leaks and OTP interception by 50%.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript
Backend: FastAPI, Django, Flask, Node.js, Express.js
Frontend: React.js, Tailwind CSS, Bootstrap, HTML, CSS
Databases: MySQL, PostgreSQL, MongoDB, Redis
Tools: Git, GitHub, AWS, Bash, PowerShell
Security: Cryptography, Authentication, HTTPS/TLS

POSITIONS OF RESPONSIBILITY

Joint-Secretary, EPMOC
IIT Una

December 2025 – Present